

19MA220/SH2220- Mathematics for Artificial Intelligence

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Started on	Monday, 23 March 2026, 11:40 AM
State	Finished
Completed on	Monday, 23 March 2026, 11:48 AM
Time taken	7 mins 16 secs
Grade	100.00 out of 100.00

Question 1

Correct

Mark 50.00 out of 50.00

Flag question

Use LU Decomposition to find L and U matrix.

For example:

Input	Result
[[3, 2, 7], [2, 3, 1], [3, 4, 1]]	[[1. 0. 0.] [1. 1. 0.] [0.66666667 0.83333333 1.]] [[3. 2. 7.] [0. 2. -6.] [0. 0. 1.33333333]]

Answer: (penalty regime: 0 %)

Reset answer

```
1 '''Program to find L and U matrix using LU decomposition.  
2 Developed by: KABILAM S  
3 RegisterNumber: 21225230119  
4 '''  
5 import os  
6 os.environ["OPENBLAS_NUM_THREADS"]="1"  
7 import numpy as np  
8 from scipy.linalg import lu  
9 A=np.array(eval(input()))  
10 P, L, U =lu(A)
```

	Input	Expected	Got	
✓	[[3, 2, 7], [2, 3, 1], [3, 4, 1]]	[[[1. 0. 0.] [1. 1. 0.] [0.66666667 0.83333333 1.]] [[3. 2. 7.] [0. 2. -6.] [0. 0. 1.33333333]]	[[[1. 0. 0.] [1. 1. 0.] [0.66666667 0.83333333 1.]] [[3. 2. 7.] [0. 2. -6.] [0. 0. 1.33333333]]	✓
✓	[[5, 1, 8], [4, 5, 7], [8, 9, 1]]	[[[1. 0. 0.] [0.625 1. 0.] [0.5 -0.10810811 1.]] [[8. 9. 1.] [0. -4.625 7.375] [0. 0. 7.2972973]]	[[[1. 0. 0.] [0.625 1. 0.] [0.5 -0.10810811 1.]] [[8. 9. 1.] [0. -4.625 7.375] [0. 0. 7.2972973]]	✓

Passed all tests! ✓

• Show/hide question author's solution (Python3)

Correct

Marks for this submission: 50.00/50.00.

Question **2**

Correct

Mark 50.00 out of 50.00

🚩 Flag question

Use LU Decomposition to solve a matrix.

For example:

Input	Result
[[3, 2, 7], [2, 3, 1], [3, 4, 1]] [4, 5, 7]	[0.875 1.125 -0.125]

Answer: (penalty regime: 0 %)

Reset answer

```
1 '''Program to solve a matrix using LU decomposition.
2 Developed by: KABILA S
3 RegisterNumber: 212225230119
4 '''
5 import os
6 os.environ["OPENBLAS_NUM_THREADS"]="1"
7 import numpy as np
8 from scipy.linalg import lu_factor,lu_solve
9 A=np.array([[3, 2, 7], [2, 3, 1], [3, 4, 1]])
10 b=np.array([4, 5, 7])
11 lu, piv = lu_factor(A)
12 x=lu_solve((lu,piv),b)
13 print(x)
14
```

	Input	Expected	Got	
✓	[[3, 2, 7], [2, 3, 1], [3, 4, 1]] [4, 5, 7]	[0.875 1.125 -0.125]	[0.875 1.125 -0.125]	✓

Passed all tests! ✓

• Show/hide question author's solution (Python3)

Correct

Marks for this submission: 50.00/50.00.